

Situated Analytics Process and Mantra

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ABSTRACT

In the current era of digitisation, a substantial amount of data significantly influences our daily experiences and decision-making processes. This data is systematically acquired from routine social interactions, online engagements, and the numerous sensors seamlessly integrated into our surroundings. Navigating this data-rich environment poses notable challenges, particularly when immersed in mixed reality environments. Situated Analytics (SA) was introduced to offer analytical reasoning within the physical space, utilizing visual analytics reasoning techniques combined with augmented reality to enhance the physical environment with real-time contextual information. The distinctive feature of SA lies in its contextual and situational, two-way, interactive visualisation, supported by a system and knowledge analytics to optimize the decision-making process. This paper discusses in-depth the components, processes, space, and mantra of SA, listing its characteristics and addressing the challenges associated with it.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models; Interaction techniques; Mixed / augmented reality; Graphical user interfaces; Interactive systems and tools; Interaction design theory, concepts and paradigms; Visualization theory, concepts and paradigms; Visualization systems and tools; Visual analytics; Visualization techniques; • **Information systems** → Spatial-temporal systems.

KEYWORDS

Situated analytics, Augmented reality, Information Visualisation, Visual analytics, Visualisation, Interaction, Interactive visualisation

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1 INTRODUCTION

The amount and complexity of data available in the physical sciences, life sciences, and social sciences are growing exponentially. Finding ways to visually translate this data will enhance our understanding of the physical environment [6]. The opportunity for new visual representations and interaction techniques is rapidly increasing with the ubiquity and stability of current advanced displays and wearable devices.

As data volumes expand, interactive visualisation tools face escalating challenges. The substantial size of diverse data imposes challenges on integration processes, necessitating the deployment of advanced and dynamic systems to meet the demands of real-time analysis. Scalability becomes crucial for handling vast datasets within constrained visual environments. To effectively address the scalability challenge, it necessitates not only keeping abreast of data growth, however also adeptly extracting insights within the limitations of confined visualisation and interaction spaces. Additionally, interaction and visualisation challenges escalate with the large amount of data streaming from wearable sensors, leading to the creation of more data than we can represent or analyse.

In 2015, Situated Analytics (SA) was introduced [7], merging augmented reality (AR) [1] and visual analytics (VA) [39]. SA provides users with analytical reasoning tools for augmented reality information visualisation, involving users in the data investigation process. The SA's context and interaction-based approaches allow a two-way dialogue between the user, the real world, and the information space, illuminating the divide between the virtual and the physical context. The analytical reasoning in SA approaches grant the connection between computer processing and human investigation. Analytical reasoning is a combination of knowledge and data analysis, and data mining approaches [10].

Given that SA originates from VA and AR, it is crucial to modify existing tools to suit the unique requirements of SA. Typically, VA tools are designed for use on standard desktop computers, equipped with a single average-sized display, keyboard, and mouse. Moreover, the AR dynamic cluttering factor inherited from the physical world background presents heightened challenges for SA. These unique challenges compel us to modify the processes and mantras of existing VA and AR to meet SA specific demands. This paper delves into

an in-depth exploration of *why SA is different?*, encompassing the challenges and requirements. Additionally, we introduce the SA mantra and illustrate its application through real-life scenarios.

2 BACKGROUND

Information visualisation's research aim is to investigate the best interaction and visualisation tools to enhance information understanding [25]. Shneiderman [35] introduced the information-seeking mantra: **Overview first, zoom and filter, then detail on demand**. This mantra divides the information visualisation operations into three main tasks: the overview representation, data manipulation, and data expansion. Each task in the information visualisation processes firmly depends on the data type and the display properties, such as the large data size and limited display space, which are the biggest challenges facing visual representation [39].

In 2005, VA [38] was introduced as a promising solution for the large data size challenge. VA combines automated analysis with interactive visualisation to extend data processing to the machine's analysis rather than relying solely on human intervention [23]. This VA human interaction involvement is used to enhance the analysis process [19] and reduce machine blindness [32]. The clustering, filtering, and sampling approaches in VA help encode the data differently, aggregating it for further human observation analyst [29]. The unique difference between the *information-seeking* and the VA mantras is that VA supports automatic analysis both before and after the interactive visualisation process, as defined by Shneiderman [35]. The mantra of VA [24] was introduced as: **Analyze First, Show the Important, Zoom/Filter, and Analyze Further - Details on Demand**.

Through the last few years, researchers have created VA analytical interaction tools, enabling users to *select, explore, filter*, and *connect* [18, 30] large datasets in a single and collaborative manner [17]. These analytical tools give users the ability to delve deeper into the presented information, independent of the data scale or complexity. With the current capabilities provided by wearable and large displays, traditional 2D interaction approaches have become largely obsolete and hinder smooth data navigation in 3D space. Ball and North [3] conducted a user study to evaluate embodied interaction for VA through large displays. They found that interaction with large displays using traditional methods was slower than embodied interaction. Users preferred physical interaction with the wall display. Accordingly, situation awareness approaches could be a potential solution for smooth and continuous VA interactions [4].

Considering that AR allows the addition of virtual information to a user's experience of the physical world, the most common form being the overlaying of computer graphics on top of a user's view of the world [1]. A major advantage of AR is in-situ presentation, allowing users to interact with a merged scene—*real* and *virtual*—in real-time [1, 16, 33]. Based on AR, SA was introduced [12] as augmented reality analytical visualisations, allowing users to manipulate and navigate through the data. SA utilizes the benefits of deploying VA to the physical scene using AR. SA's contextual engagement increase information understanding by registering data on its related physical node [9]. Nonetheless, due to the spatial

and contextual filter characteristics of AR resulting from the surrounding physical environment, the existing VA mantra cannot be directly applied to SA.

The persistent challenge of limited display inherited from AR remains a focal point for SA [26, 40]. Over recent years, researchers have explored diverse visualisation techniques to incorporate information within AR environments [20, 36, 37]. Kalkofen et al. [21] categorises AR visualisation techniques into three principal approaches: *Data integration*, *scene manipulation*, and *context-driven* visualisation. While these techniques aim to reduce clutter in AR scenes, they inherently limit the amount of data presented to users, giving rise to the "Knowledge-Blinded" problem [7].

To mitigate data filtering and reduce cluttering, researchers have investigated approaches such as blending information with real-world objects [22], employing diminished reality [28], utilizing color corrections [14, 15], and employing context-driven visualisation [40]. Although these blended visualisation techniques alleviate cluttering in situated visualisation scenarios [27, 31], they do not entirely resolve clutter for abstract data represented in the view. This is attributed to the inability to remove the physical context from the scene, thereby limiting the representation canvas for abstract visualisation.

Recently, researchers explored the integration of abstract visualisation into physical environments. However, the challenge is significantly amplified by the rising volume of abstract data [2, 9, 11]. Moreover, with the requirement for both situated and abstract visualisation, which necessitates the manipulation of AR and analytical interaction for SA process, we will need dynamic and generic User Interfaces (UI) [5, 9] to deploy SA in real-life scenarios [8], which remains one of the limitations of SA analytical interfaces [34].

3 SITUATED ANALYTICS REQUIREMENTS AND CHALLENGES

Generating and augmenting data visually into the real scene entails unique requirements and advantages, stemming from AR technology. SA visualisations represent both data and relationships between nodes. They must facilitate data aggregation, encoding, and modification of the visual representation based on contextual and situational parameters. Thus, SA should facilitate interactive visualisation tasks, including **overview, zoom/filter, and details on demand**.

In a shopping scenario, envision a user navigating between supermarket aisles equipped with a Head-Mounted Display (HMD) AR device. The AR display seamlessly integrates SA visualisations into the real scene, allowing the user to perceive computer-generated information associated with actual objects. Figure 1 illustrates the three visualisation tasks. In **overview** mode, the user *selects* a milk carton from the supermarket shelf. Following this initial selection, the system color-codes the physical milk, indicating its compatibility with user preferences, and presents a detailed abstract view in the top left corner. Additionally, the system augment the remaining shelf products with virtual health bars, illustrating their compatibility percentage with the chosen milk carton (see Figure 1a). The compatibility health bars are color-coded to represent the nutritional impact of each product. Building on the overview representation, the user decides to *explore* more detailed information about

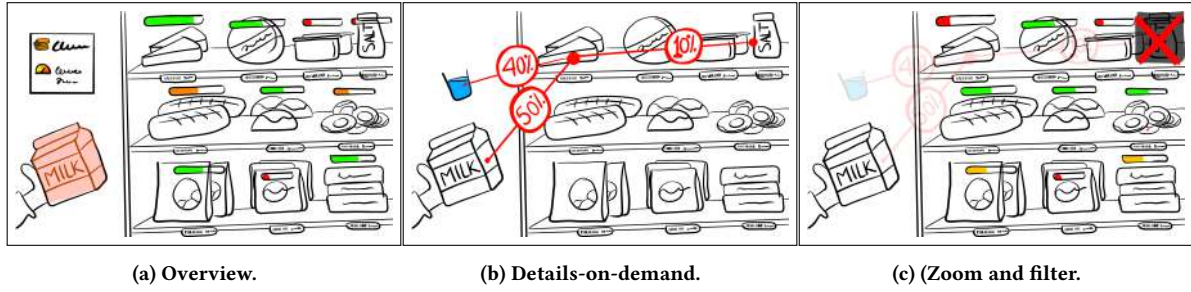


Figure 1: Information seeking tasks using situated analytics.

the most compatible product, which is the cheese. This **details on demand** view showcases the ingredients of the cheese block and identifies other products on the shelf sharing common ingredients with the selected milk carton (see Figure 1b). The system displays a numerical percentage of each ingredient on the connecting relationship lines. Following this detailed exploration, the user discovers that the chosen cheese has a high sodium content. Based on this observation, the user adjusts the query parameters **analyse** by **filtering** the sodium content. This reconfiguration modifies the visual representation based on the low sodium requirement (see Figure 1c).

Considering relationship representation as one of the primary components in SA visualisation approaches, it can take the form of either implicit or explicit methods. Implicit approaches alter the visual display based on the interrelationships between real objects and the overarching goal. An example of this implicit approach is the modification of the *life bars* linked to physical objects in response to user interaction and a predefined calorie budget. For instance, the *life bar* for chips appeared green (see Figure 1a) but changed to a reddish color after the user's salt filter request (see Figure 1c). This implicit visualisation enables users to comprehend results through **clustering**, **comparing**, or **sorting**. On the other hand, the explicit visualisation approach portrays the relationship between physical objects and their associated data, exemplified by connecting lines between the selected product on the shelf and other related products. This relationship-based visualisation offers users a customized, informative, and detailed response, moving beyond a binary yes or no. In the aforementioned shopping scenario, the AR space encounters more challenges (Figure 2) compared to traditional approaches, including:

- **Cluttered background**—when using an AR display, the background for data representation is the real scene, introducing additional cluttering factors (see Figure 2d).
- **Off-screen objects**—in SA, the physical world serves as the user's analytical interface; however, the real scene extends beyond the representation canvas, causing most interactive nodes to be out of the AR display's field of view (see Figure 2e).
- **Limited display area**—the cluttered background challenge results in very few remaining areas for the presentation of abstract information (see Figure 2f).

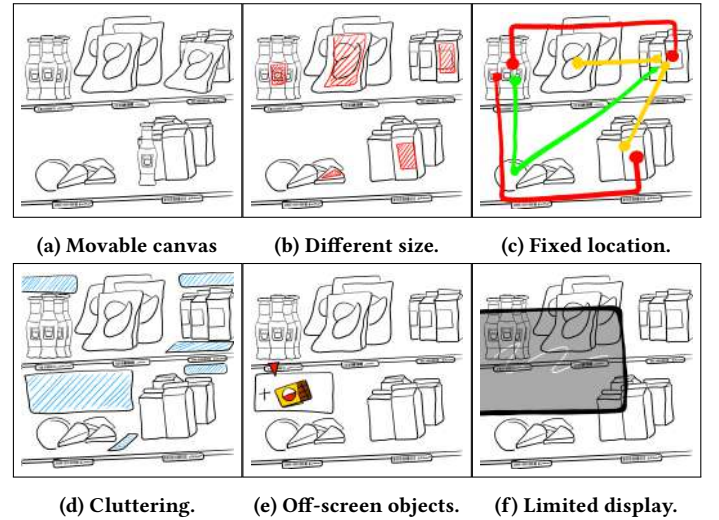


Figure 2: Situated analytics visualisation challenges.

- **Movable canvas**—where the canvas areas in situated visualisations are the surfaces of movable real objects (see Figure 2a).
- **Different canvas sizes**—as the canvas size in situated visualisations is determined by the real object's surface size (see Figure 2b).
- **Fixed location**—displacement, a commonly used approach in information visualisations, is challenging in SA visualisations as the data nodes in SA are the physical objects in the real scene that cannot be virtually re-arranged (see Figure 2c).

However, with the mentioned limitation of deploying SA, it should not only support visual representations for analytical tasks but also facilitate interaction. This would enable users to perform visual analytics in a physical environment, with the benefit of exploring and analysing data interactively, such as:

- **Allow relationship visualisation**—represent the data associated with individual physical objects and the relationship between distinct physical objects (see Figure 3a).
- **Ability to explore detailed information**—support multi-depth data representation, allowing the exploration of data

associated with physical objects through user interaction (see Figure 3b).

- **A visual representation for data aggregation operations**—SA should allow the users to perform multiple queries on the same scene, such as sort (see Figure 3c), zoom & filter (see Figure 3d), and cluster (see Figure 3e).
- **Support zooming and off-screen**—enabling users to view and interact with the physical objects that are outside the AR display's Field Of View (FOV) (see Figure 3f)

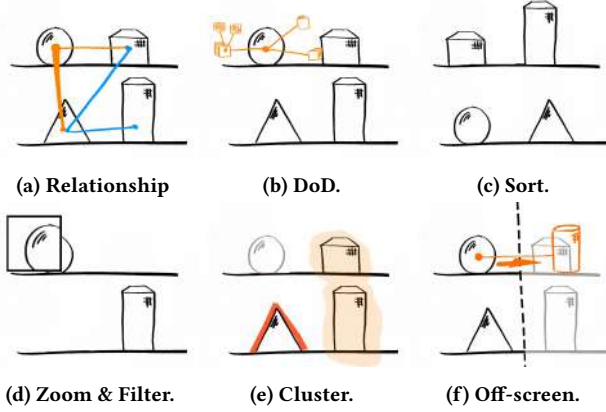


Figure 3: Situated analytics data aggregation.

4 SITUATED ANALYTICS COMPONENTS AND SPACE

The primary objective of SA is to empower users in making informed decisions by visually reflecting the outcomes of their interactions [7]. SA enables users to engage in immersive analytical tasks within the physical world, allowing interactions with both physical objects—referred to as *Physical space*—and information attached to each object—referred to as *Information space*. These interactions empower users to execute complex queries within the context of a physical object and visualise the results of the applied query.

Each SA system comprises four UI components [13]: situated representation, abstract representation, AR interaction, and analytical interactions. *Situated representation* augment information to physical objects, based on user analysis. *Abstract representation* offers a summary and outcome of the analytical interaction between users, physical objects, and attached information, without spatial correlation to the physical scene. *AR interactions* empower users to engage with physical objects within a query context, involving actions like moving, selecting, or removing an object in the physical world. Meanwhile, *Analytical interactions* enable users to interact with information related to physical objects through analytical queries, encompassing actions such as filtering, sorting, and comparing.

In light of the SA space introduced by ElSayed et al. [10], each of the four SA UI components is expected to reside within the *SA cubic space*. This space is defined by three axes: the *visualisation axis* (X_{SA}), ranging from abstract overlay representation to situated

blended representation (horizontal axis); the *interaction axis* (Y_{SA}), transitioning from discrete interaction to continuous interaction (vertical axis); and the *working immersion axis* (Z_{SA}), progressing from working in the information space to the physical space (depth axis). Consequently, the deployment of one session of the SA shopping analysis task, as mentioned in (Figure 1), will result in different SA dimension states. For example:

- **State 1 (●):** The user navigates between supermarket aisles, views an overview of information on products, and then engages with *situated* visualisation (X_{SA}). The analysis is altered *continuously* (Y_{SA}) by walking in the *physical* space (Z_{SA}).
- **State 2 (■):** The user selects a product through *discrete* action (Y_{SA}), views a color-coded *blended* visualisation showing the compatibility of the product with user preferences and budget (X_{SA}). This analysis is performed by holding the *physical* product box (Z_{SA}).
- **State 3 (▲):** The user examines the reason for the color-coded value by looking at the *abstract* representation *overlaid* in the top left corner of the view (X_{SA}). *No* additional *interaction* is required for the analysis (Y_{SA}), and the user views the presented information with *no* correlation to the physical environment (Z_{SA}).

Figure 4 illustrates the challenges in developing a SA system, as users need to transition between various visualisation and interaction modalities in different immersive views. A proficient SA system should enable seamless transitions between different modalities, empowering users to concentrate on analysis and sense-making with minimal technological distractions or comprehension challenges.

In the following section, we delve into the SA process and its mantra, presenting the state diagram and factors influencing the SA system. This exploration requires consideration by those involved in constructing SA systems, providing a comprehensive overview of the theoretical framework governing the flow between analytical steps. It takes into account the influential role of the SA space in shaping this workflow.

5 SITUATED ANALYTICS PROCESS

Inspired by VA, SA are analytical reasoning techniques, enabling users to interact with and visualise data in the two merged spaces—the real world and the virtual space. SA techniques influenced by the user's behavior and interaction history. Moreover, SA is a relationship-based representation, revealing both explicit and implicit relationships between data nodes. Figure 5 shows the adapted process model of SA based on the VA model. Three extra node were added, showing the new SA components: *B*: the blended view, *P*: the physical data set, and *A*: the virtual data set, which is associated to the physical objects. Figure 5 shows the adapted process model of SA as a transformation $F : S \rightarrow I$, where:

- *S* is the data sets $S = \{ \{P_1, \dots, P_m\}, \{A_1, \dots, A_n\} \}$, where each set of physical objects $P_i, i \in (1, \dots, j)$ is associated to a set of virtual content $A_{i1}, \dots, A_{iK}$, and modifications to the data of either set (*P* or *A*) can occur independently, without necessarily affecting the other. Modifications to the virtual content *A* are enabled through SA authoring tools,

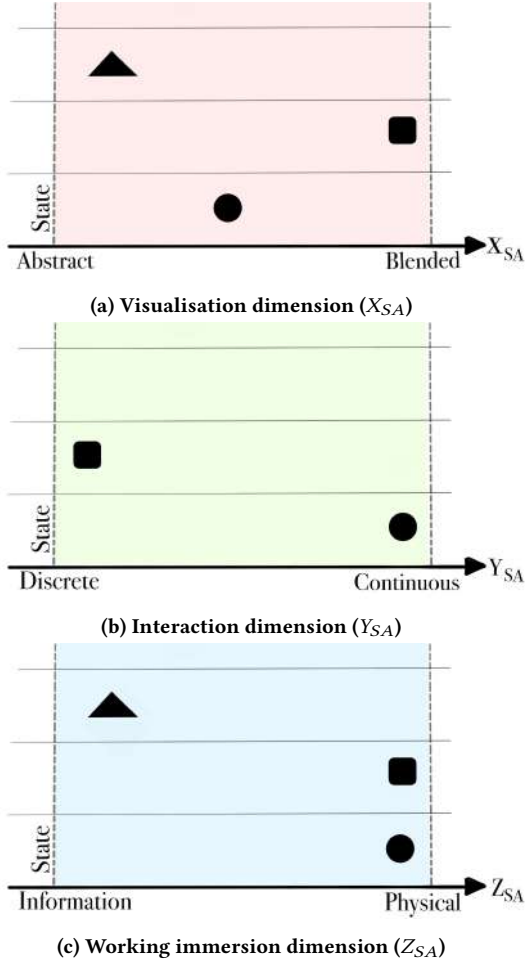


Figure 4: The deployment of SA task in the SA space dimensions [10]

while interactions with the physical environment may result in alterations to the associated set of physical objects P .

- I is the output insight, that can be obtained from the data set S , either through the created visualisation B or through confirmation of hypotheses H . The insight I cannot be obtained from the created visualisation directly.
- F is a concatenation of functions $f \in \{D_W, V_X, B_{X'}, H_Y, U_z\}$ defined as follows:
 - D_W is the basic data pre-processing, with $D_W : S \rightarrow S$, and $W \in \{T, C, SL, I, CX\}$, D_T is the data transformation functions, D_C is the data cleaning functions, D_{SL} is data selection functions, and D_I is the data integration functions. D_{CX} is data physical context transformation functions such as: contextual, spatial, situation filtering.
 - V_X is the abstract visualisation generation functions, with $X \in \{S, H\}$, generate a virtual visualisation either through data $V_S : S \rightarrow V$ or through hypotheses $V_H : H \rightarrow V$, or through the physical scene $V_B : B \rightarrow V$

- $B_{X'}$ is the created blended visualisation, with $X' \in \{V, H\}$, visualising data either through created virtual visualisation combined with the physical scene $B_V : V \rightarrow B$ or through hypotheses $B_H : H \rightarrow B$
- H_Y is the hypothesis generation process, with $Y \in \{S, V\}$, generating the hypotheses either through data $H_S : S \rightarrow H$ or through the created visualisation $H_V : V \rightarrow H$
- U_Z is the user interaction, with $Z \in \{V, H, B, CB, CH\}$. User interaction can effect only the created virtual visualisation $U_V : V \rightarrow V$, or the blended scene—the virtual visualisation merged in the physical scene— $U_B : B \rightarrow B$ or effect only the hypotheses $U_H : H \rightarrow H$. User interaction can conclude the insight I from either the blended view $U_{CB} : B \rightarrow I$ or from the hypothesis $U_{CH} : H \rightarrow I$.

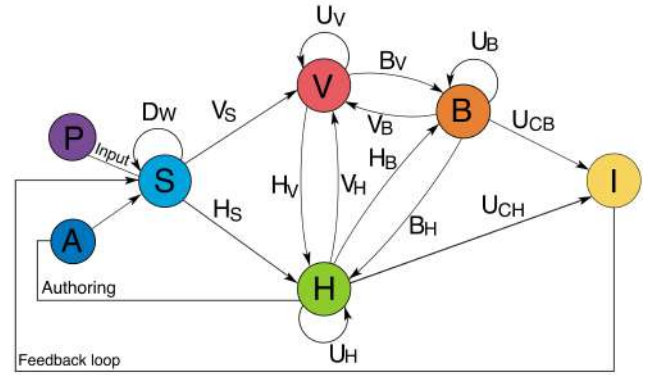


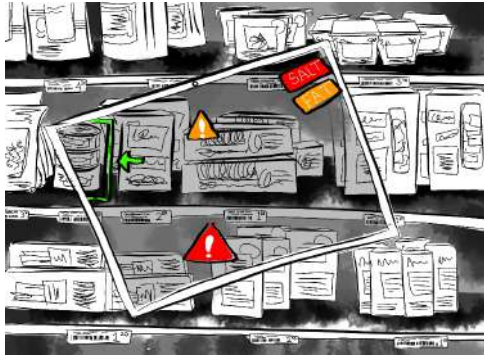
Figure 5: The SA process, which is adapted from the formal VA description model [24], where: dataset S consist of physical objects P and associated data A . Users can hypothesise using the dataset S , the abstract visualisation V , or the blended visualisation B , concluding the output insight I

6 SITUATED ANALYTICS USE CASE

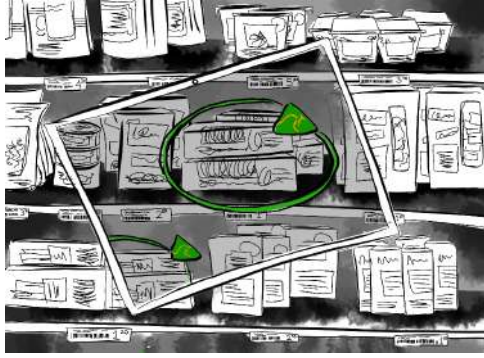
Let's reconsider the SA personal assistant in a shopping context, where the system could provide interactive decision-making aids. Users can interact (**User Interaction** U) with the supermarket products, representing data nodes (**Data set** S). These data nodes can be the product boxes (**Physical objects** P) or the information associated with them (**Virtual content** A). Such an application would allow users to query data through a **Concatenation of Functions** F , such as compare, sort, filter, and find, based on users' hypotheses (**Hypothesis Generation Process** H).

Based on users' queries, the output information is presented to users (**Output Insight** I) as either an AR overlay **Visualisation** V or a **Blended Visualisation** B (see Figure 6a). However, the insight I cannot be obtained directly from the created visualisation, as it is filtered by the situational and contextual parameters (**Data physical context transformation** D_{CX}) of the SA system, such as the user's location in the supermarket, products in the view, and purchase history.

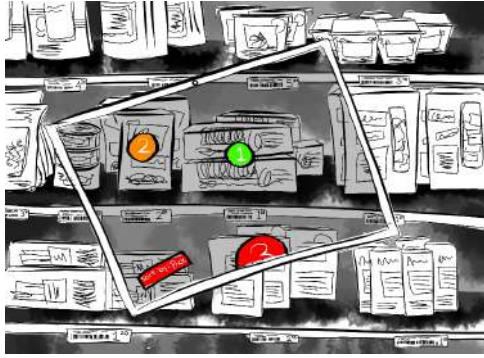
Figure 6 shows a **hypotheses** scenario (H), where a user is trying to choose a breakfast meal through real-time interaction (**User**



(a) Hypothesis generation process (H) based on user's goals.



(b) Output insights (I) presented as visualisation (V) blended annotations (B).



(c) Analyse further using concatenation of functions F .

Figure 6: SA personal assistant analytical process.

Interaction U_H with the physical products (**Physical objects P_i**) and their associated data (**Virtual contents A_i**). This hypothesises analytical query initially builds on stored data set (H_S) such as: the user profile, which holds the nutrition functions, colour and cost budget, fitness goals, and constraints.

Based on the (**Output Insight I**) the user hypothesises the insight for a further step H_V (See Figure 6b). The user filters the shelf products based on the locally made parameter, leading to highlighting of the query results. The user filter products by focusing on the interested physical objects (U_B). Then, the user starts interacting with the

visualisation (U_V) (see Figure 6c). Based on the (**Output Insight I**), the user hypothesises for a further step H_V (See Figure 6b). The user filters the shelf products based on locally set parameters, leading to the highlighting of the query results. The user filters products by focusing on the interested physical objects (U_B). Subsequently, the user starts interacting with the visualisation (U_V) (see Figure 6c).

SA demands a careful equilibrium between spatial and contextual factors to enable a fair comparison with contextual influences, ensuring the effective presentation of all essential SA data nodes. Unlike the VA mantra: "**Analyse first - Show the Important - Zoom, Filter and Analyse Further - Details on Demand**", SA frame's the user's initial view based on the physical displayed zone. Then, users can analyse—the physical scene and the associated data—through the blended view (as demonstrated in Figure 5), investigating the contextual relationship on demand. These process differences make SA almost the opposite of the traditional mindset, resulting in: "**Details First, Analysis, then Context on Demand**" [12].

7 CONCLUSION

SA facilitates user interactions with physical objects and the manipulation of associated data, supporting users in decision-making through AR-presented multi-dimensional data and enhancing analytical abilities. This paper offers a comprehensive exploration of SA, encompassing its components, process, and mantra. The paper delves into key challenges and limitations of situated analytics in real-life scenarios. We introduced an adaptive SA mantra aligned with the VA mantra, taking into consideration parameters that influence the SA process.

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A APPENDIX: SITUATED ANALYTICS APPLICATIONS

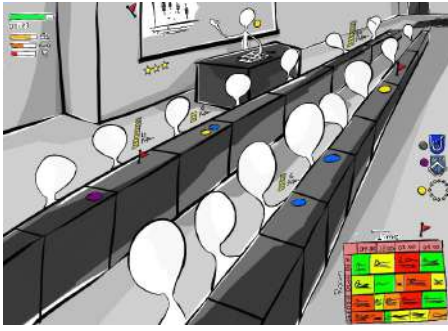
The appendix presents detailed exemplars illustrating the tangible use cases of SA. These examples explore the details of SA functionality, explaining SA main features and listing the benefits it provides across different situations and contexts. Furthermore, the appendix outlines the systematic deployment of our mantra in each scenario, providing a step-by-step account of its implementation.

A.1 Social Networking Applications

Figure 7 shows a use case of SA in a conference room scenario. The user sees the room through an AR display, providing them with more information associated (**A**) with objects in the physical world (**P**). The user can see the information (**I**) as blended view (**B**) to the physical scene, such as: a star visualisation registered to the presentation stage that shows the real-time rating of the presentation. This registered overall information allows the user to see the audience's current interest in the talk. At the bottom right of the AR view, the user can visualise abstract information (**V**) such as, that has been automatically proceed (**D**): the conference schedule, which is colour-coded based on the user's preference. On the opposite side of the view—the top left corner—the user can see their daily activities, based on their schedule and predefined goals.

Figure 7a shows a blended visualisation (**B**) registered to each audience member, representing the audience's research relationship with the talk topic by using bar charts for the number of the published papers; combined with a colour-coded key to cluster them based on their institutes. The user then starts to interact with the scene (**U**). The user add a flag to the audience members (U_B), reminding themselves to speak with them later. Figure 7b illustrates an outcome of a user query. The user filters the audience members based on two keywords from a paper they is reading. This query (**H**) alters the blended visual representation (**B**), adding colour-coded arrows pointing to the audience members who are working in the mentioned fields. The visualisation also shows a "running" annotation associated with one of the audience members, as this specific researcher is leaving soon, based on the their stored schedule.

SA provides an open two-way interactive session between users and the real and virtual spaces, allowing them to investigate continuously (**H**) through data processing (**D**); for example, sorting (D_{SL}) the audience based on the user's research preference, or view the data in the user's view (D_{CX}) (see Figure 7b). Based on this further



(a) An SA visualisation in a conference room. Annotation and colour-coding represent information about the audience members and the session talk.



(b) The user filter the AR scene based on the research keyword on the paper they are reading.



(c) The user rank the conference's members based on the user's research interest.

Figure 7: SA Social networking use case application.

investigation (F), the visualisation registers a ranking value onto each one of the researchers (U_B), with colour-coding to enhance the visualisation's visibility amongst the real world clutter big space.

A.2 Navigations Applications

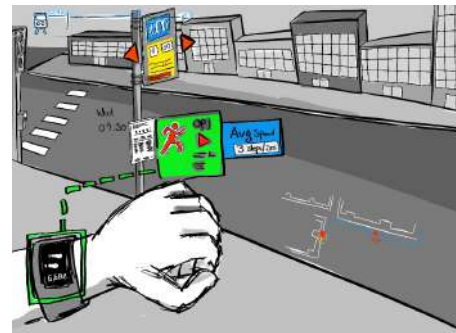
Navigation is also another potential situation for SA application. The increased enhancement of ubiquitous (pervasive) and wearable computers makes navigation a promising area for situational awareness decision making. Figure 8 illustrates the scenario of trip planning. The existing wearable devices such as mobile phones,

fitness bands, and smart watches feeds real-time data (S) that can enhance the situational understanding. Figure 8a shows the presented overall information, based on the entered destination.

Looking through an AR display, the user can see the current transportation information registered (B) on the bus station's sign. Moreover, a GPS tracking for the bus is registered (B) on the street. Based on the system recommendations in Figure 8b and the filtered data presented (D), the user notices that they have two options (H): either wait 9.5 minutes for the next bus or run towards another station. The "running" annotation is visualised in a red colour, indicating a warning associated with the second option.



(a) The user investigate a transportation options. The user has two options either wait for the next bus or run to a near bus station.



(b) The user views details break down of the presented option, syncing the fitness information and goals.



(c) Selecting the running option will burn calories, enabling the user to have the cookies they stored on their "wish list".

Figure 8: SA navigations applications.

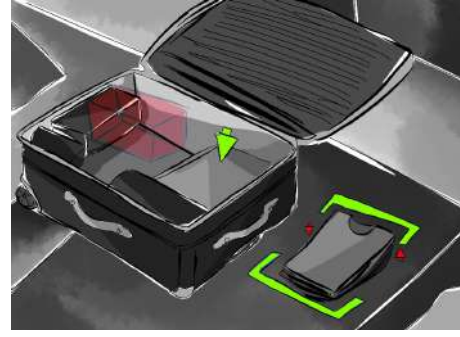
The user then expands the "running" annotation (U_H), showing that the warning reason shows a fast running speed, which is faster than the user's average running speed, based on the fitness band D_C . This running option, however, will increase the amount of calorie burning. Based on the presented information (U_{CB}) (see Figure 8c) the user manipulating the query parameters by pushing up the priority of the fitness (U_B and U_H). So, the system advises the user to run for ten minutes east toward the second station blended to the situated visualisation (B). The system also marks a supermarket on the visualised map as an abstract visualisation (B) where the user interact (U) can pick one of their wish list's candies because of the amount of consumed calories over the running distance calculated and represented (U_{CB}) to him.

A.3 Real-world Assembly Applications

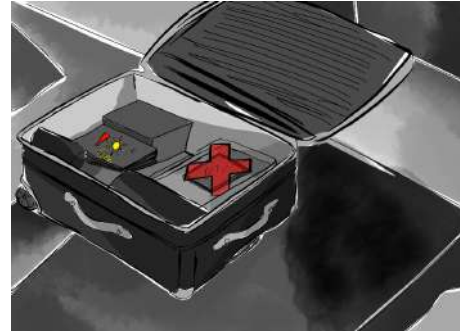
Real object assembly is a well-known goal that is being investigated for AR. The existing approaches, however, are based on the spatial properties of the physical objects, without involving the information associated with the physical objects.

Figure 9 shows a knowledge awareness example, for luggage packing based on automated data pre-processing (D). The user can see a recommended order (D_{SL}) and assembly for the bag (B), using navigation arrows (see Figure 9a). The user starts to assemble the bag (U_{CB}) based on the blended view (physical bag and recommending visual cues) in front of him. These virtual annotations are based on the user's presorted trip schedule (D). Knowing from the schedule that the meeting will be held on the second day of the trip (H), the system recommends that casual wear will be worn first and, thus, should be packed last. Figure 9b shows that the system warns the user against putting the tablet in the suitcase using situated visualisation (B), as they allocated a reading task on the flight so will need the tablet in their backpack. The user fixed the status based on the presented information (I), by removing the tablet from the bag (U_I).

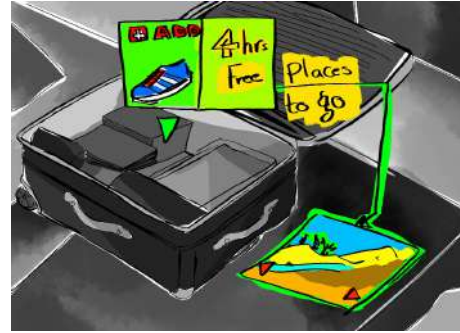
Also, the SA registers a warning annotation on a jacket inside the bag, informing the user about their destination temperature 43 degrees as an abstract presented visualisation (V). So the user can consider removing the jacket from the bag (U_I). Finally, the system suggests to the user adding sports shoes to be used in their trip'd free hours. The SA system supported their suggestion by "places to go" recommendations, based on the user's schedule (V), interest and goals (as demonstrated in Figure 9c). The user then order (U_H) with these "places to go" based on his interest, altering the information presented and the bag assembly recommendations.



(a) The user arranges the bag based on the trip schedule and the spatial areas. The system recommends to change the clothes' order to match the schedule.



(b) Annotations are registered on the bag content, altering the user to remove the tablet and concern replace the heavy clothes based on the weather information.



(c) The SA system provide the user with system recommendations.

Figure 9: SA real-world assembly applications.